Supplementary Materials

This supplementary file contains the original supplementary tables and figures, together with the additional content moved from the main manuscript during proof revision for page reduction, in order to meet the journal's maximum 14-page limit for a Regular Paper.

Supplementary Text S1. Computational Performance and Scalability

Finally, we assessed the computational cost of SPSID. The $O(n^3)$ complexity, which is standard for methods involving matrix inversion, arises from the single inversion step in Equation 10. We benchmarked the wall-clock runtime of all denoising methods on synthetic networks of increasing size n (see Supplementary Table S1 for full results). The results in Supplementary Table S1 show that SPSID's absolute runtime remains practical for typical GRN sizes. On a standard desktop, SPSID takes on average 0.24 s (SD 0.02) at $n = 500$, 1.01 s (SD 0.02) at $n = 1000$, 5.33 s (SD 0.01) at $n = 2000$, 17.49 s (SD 0.17) at $n = 3000$, 48.47 s (SD 0.06) at $n = 4000$, and 148.72 s (SD 0.25) at $n = 6000$. SPSID is consistently faster than RENDOR at all sizes, and both are *orders of magnitude* faster than ND (e.g., ND requires 2373.05 s at $n = 6000$). Simpler baselines such as Silencer and ICM are the fastest in wall-clock time, which reflects their much cheaper update rules but also their lower accuracy. NE lies between SPSID/RENDOR and these low-cost baselines. Overall, SPSID offers competitive runtime among the higher-accuracy denoisers: its cubic cost matches that of other global diffusion-based methods, yet remains tractable for networks up to $n = 6000$.

Table S1: Runtime comparison on synthetic networks of increasing size n.

Method \ n	500	1000	2000	3000	4000	5000	6000
SPSID	0.24 ± 0.02	1.01 ± 0.02	5.33 ± 0.01	17.49 ± 0.17	48.47 ± 0.06	89.67 ± 0.14	148.72 ± 0.25
RENDOR	0.14 ± 0.03	0.53 ± 0.01	5.04 ± 0.16	18.48 ± 0.55	53.94 ± 0.21	102.03 ± 0.41	174.87 ± 1.24
ND	0.42 ± 0.00	3.38 ± 0.04	53.70 ± 0.32	246.55 ± 0.73	651.77 ± 1.24	1298.00 ± 2.70	2373.05 ± 1.05
NE	0.11 ± 0.00	0.48 ± 0.03	2.53 ± 0.02	10.38 ± 0.14	23.06 ± 0.62	42.23 ± 0.29	71.24 ± 0.16
Silencer	0.03 ± 0.01	0.17 ± 0.03	0.82 ± 0.02	3.07 ± 0.12	6.09 ± 0.14	10.59 ± 0.13	16.92 ± 0.23
ICM	0.03 ± 0.00	0.17 ± 0.07	0.58 ± 0.03	2.09 ± 0.17	3.86 ± 0.17	6.26 ± 0.06	10.16 ± 0.06

Notes: Entries report mean ± standard deviation of wall-clock time (in seconds) across repeated runs on the same hardware.

Table S2: Mean AUROC / AUPR under varying sparsity (ρ), noise (σ) and transitivity (β).

(a) Varying network density ($\sigma = 0.5,\ \beta = 0.5$)

Method	$\rho = 0.01$		$\rho = 0.03$		$\rho = 0.05$		$\rho = 0.07$		$\rho = 0.09$	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
SPSID	**0.909**	**0.177**	**0.896**	**0.277**	**0.878**	**0.305**	**0.856**	**0.313**	**0.832**	**0.317**
RENDOR	0.865	0.096	0.842	0.170	0.816	0.203	0.789	0.222	0.764	0.238
ND	0.857	0.105	0.835	0.164	0.807	0.187	0.778	0.201	0.751	0.214
NE	0.830	0.089	0.810	0.168	0.777	0.199	0.744	0.216	0.717	0.230
Silencer	0.526	0.012	0.521	0.033	0.517	0.054	0.514	0.074	0.511	0.094
ICM	0.453	0.009	0.460	0.027	0.467	0.046	0.472	0.064	0.476	0.084

(b) Varying noise level ($\rho = 0.05,\ \beta = 0.5$)

Method	$\sigma = 0.1$		$\sigma = 0.3$		$\sigma = 0.5$		$\sigma = 0.7$		$\sigma = 0.9$	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
SPSID	**0.957**	**0.528**	**0.940**	**0.454**	**0.878**	**0.303**	**0.811**	**0.208**	**0.758**	**0.156**
RENDOR	0.939	0.379	0.894	0.296	0.816	0.202	0.750	0.146	0.703	0.116
ND	0.925	0.332	0.887	0.270	0.807	0.187	0.740	0.135	0.693	0.108
NE	0.884	0.311	0.853	0.278	0.777	0.199	0.715	0.144	0.671	0.114
Silencer	0.537	0.059	0.523	0.055	0.516	0.054	0.514	0.053	0.511	0.052
ICM	0.439	0.042	0.455	0.044	0.466	0.045	0.473	0.046	0.478	0.047

(c) Varying transitive strength ($\rho = 0.05,\ \sigma = 0.5$)

Method	$\beta = 0.1$		$\beta = 0.3$		$\beta = 0.5$		$\beta = 0.7$		$\beta = 0.9$	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
SPSID	**0.909**	**0.446**	**0.897**	**0.381**	**0.879**	**0.304**	**0.858**	**0.237**	**0.837**	**0.188**
RENDOR	0.855	0.269	0.839	0.240	0.816	0.201	0.792	0.166	0.765	0.136
ND	0.847	0.251	0.831	0.226	0.807	0.187	0.780	0.150	0.752	0.123
NE	0.804	0.245	0.796	0.227	0.777	0.199	0.754	0.168	0.725	0.141
Silencer	0.525	0.054	0.522	0.054	0.518	0.054	0.513	0.053	0.510	0.053
ICM	0.460	0.045	0.460	0.045	0.466	0.045	0.471	0.046	0.478	0.047

Note. Bold values indicate the best results. Methods shown are SPSID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).
This table is cited in the main text to report IV-C. Extended-Scenario Analysis.

Table S3: Mean AUROC / AUPR on G2 simulation model at boundary condition ($\beta_1 = 1.0$).

Boundary condition simulation ($\rho = 0.05,\ \sigma = 0.5,\ \beta_1 = 1.0$)

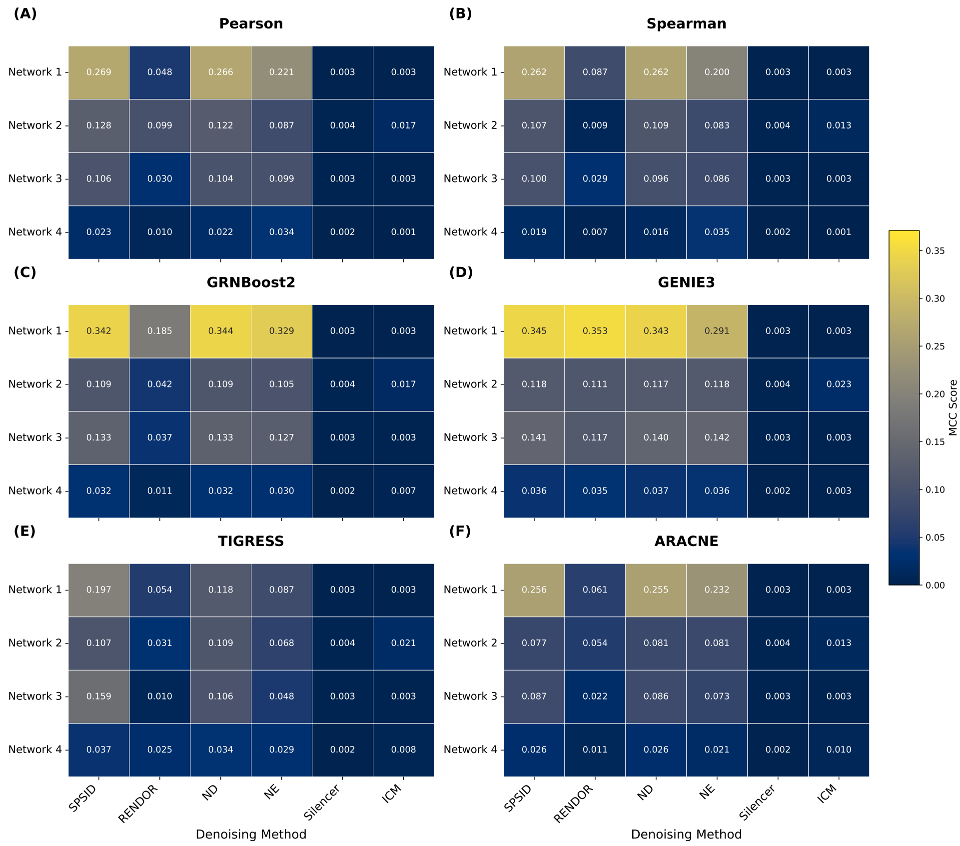
Method	AUROC	AUPR
SPSID	**0.828**	**0.172**
RENDOR	0.754	0.125
ND	0.740	0.113
NE	0.714	0.130
Silencer	0.508	0.052
ICM	0.481	0.047

Note. Bold values indicate the best results. Methods shown are SPSID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).
This table is cited in the main text to report IV-C. Extended-Scenario Analysis.

Table S4: Mean AUROC / AUPR on higher-order simulation model.

Varying cubic (G3) transitive strength (β_2) ($\rho = 0.05,\ \sigma = 0.5,\ \beta_1 = 0.5$)										
Method	$\beta_2 = 0.1$		$\beta_2 = 0.3$		$\beta_2 = 0.5$		$\beta_2 = 0.7$		$\beta_2 = 0.9$	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
SPSID	**0.869**	**0.271**	**0.829**	**0.202**	**0.788**	**0.157**	**0.757**	**0.131**	**0.733**	**0.115**
RENDOR	0.814	0.201	0.785	0.178	0.750	0.147	0.719	0.123	0.694	0.108
ND	0.805	0.184	0.775	0.152	0.736	0.123	0.705	0.105	0.679	0.094
NE	0.764	0.189	0.727	0.158	0.693	0.133	0.667	0.115	0.649	0.103
Silencer	0.517	0.054	0.512	0.054	0.508	0.053	0.506	0.053	0.505	0.052
ICM	0.467	0.045	0.475	0.046	0.480	0.047	0.485	0.047	0.487	0.048

Note. Bold values indicate the best results. Methods shown are SPSID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).
This table is cited in the main text to report IV-C. Extended-Scenario Analysis.



Supplementary Fig. S1: DREAM5 MCC benchmark performance. Facet heatmap displaying MCC scores. Each panel (A–F) represents one of the 6 upstream inference methods. Columns compare SPSID against 5 other denoising baselines across the 4 DREAM5 networks. This figure is cited in the main text to report V-B. Overall Performance.

Supplementary Text S2. Rank-score Comparison Across Methods

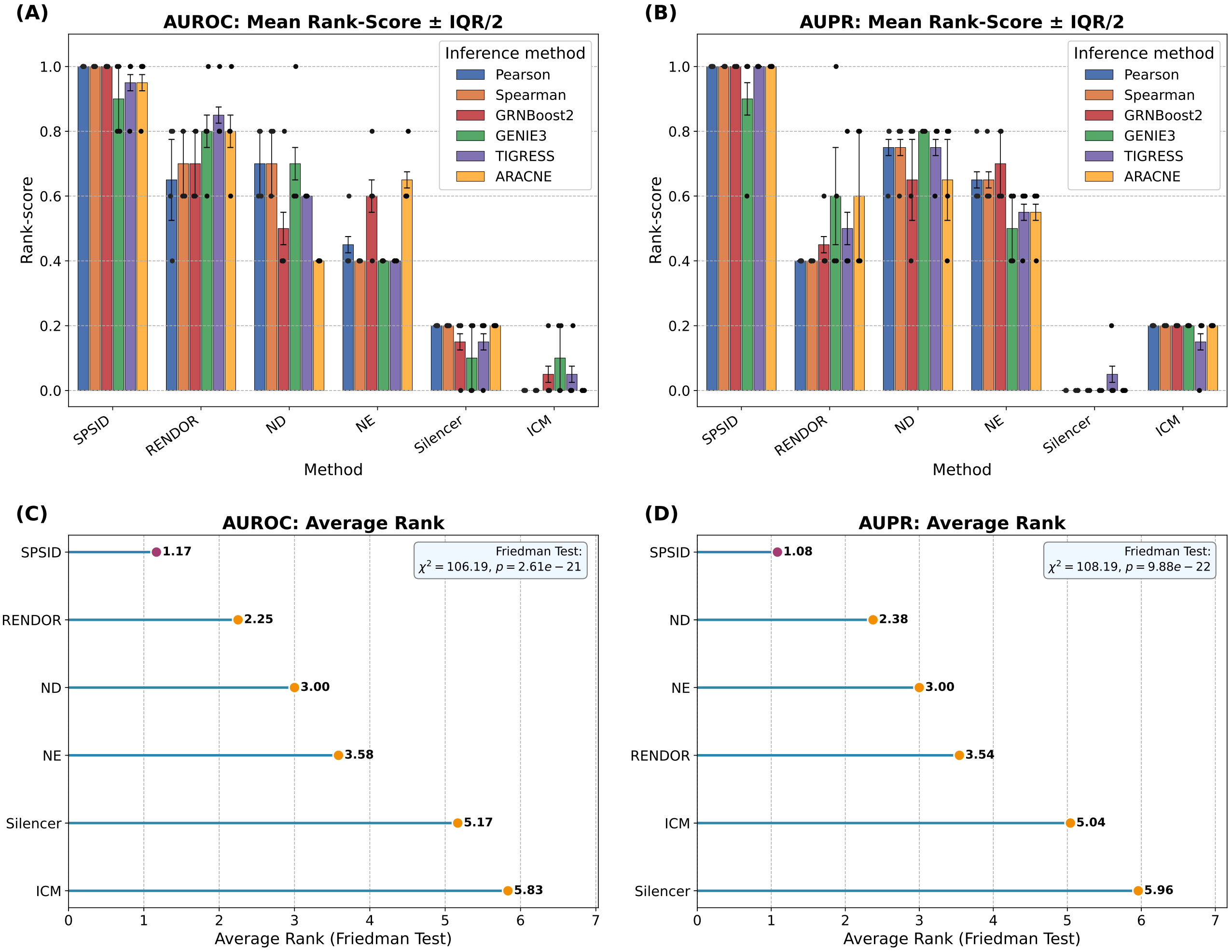
To assess performance consistency beyond absolute scores, we adopted a rank-based evaluation. For each of the 24 DREAM5 combinations (4 networks × 6 upstream inference methods), we ranked the M competing methods based on their AUROC and AUPR scores.

The rank $r_i \in \{1, \ldots, M\}$ for a denoising method i on a given task is converted to a Rank-score (s_i) as follows:

$$s_i = 1 - \frac{r_i - 1}{M - 1}, \qquad s_i \in [0, 1].$$

This score normalises the rank to the $[0, 1]$ interval, where the top-performing method receives a score of 1 and the worst receives a score of 0. This approach allows for robust aggregation of relative performance across different tasks.

Supplementary Fig. S2 provides a comprehensive analysis of these ranks. Panels A and B show that SPSID attains nearly perfect mean rank-scores in both metrics and displays the narrowest interquartile bands, confirming that it never falls below second place for any network–inference combination. RENDOR and ND form a middle tier whose mean scores are 0.15–0.25 points lower and whose variability is appreciably larger, indicating sensitivity to the choice of upstream engine. Network Enhancement trails this pair, whereas Silencer and ICM cluster close to zero, often occupying the last rank when precision-recall performance is considered. Panels C and D provide the Friedman average ranks: SPSID achieves a near-perfect 1.17 for AUROC and 1.08 for AUPR, versus 2.25 (RENDOR) and 2.38 (ND) for the nearest competitors in each metric, respectively. The overall Friedman statistics are highly significant ($\chi^2 = 106.19$ for AUROC, $\chi^2 = 108.19$ for AUPR), corresponding to $p = 2.61 \times 10^{-21}$ and $p = 9.88 \times 10^{-22}$, respectively. These rank-based assessments, which are independent of absolute metric magnitude, reinforce the earlier percentage-gain results: SPSID not only delivers good average accuracy but also consistent behaviour across disparate data sets and upstream inference pipelines.

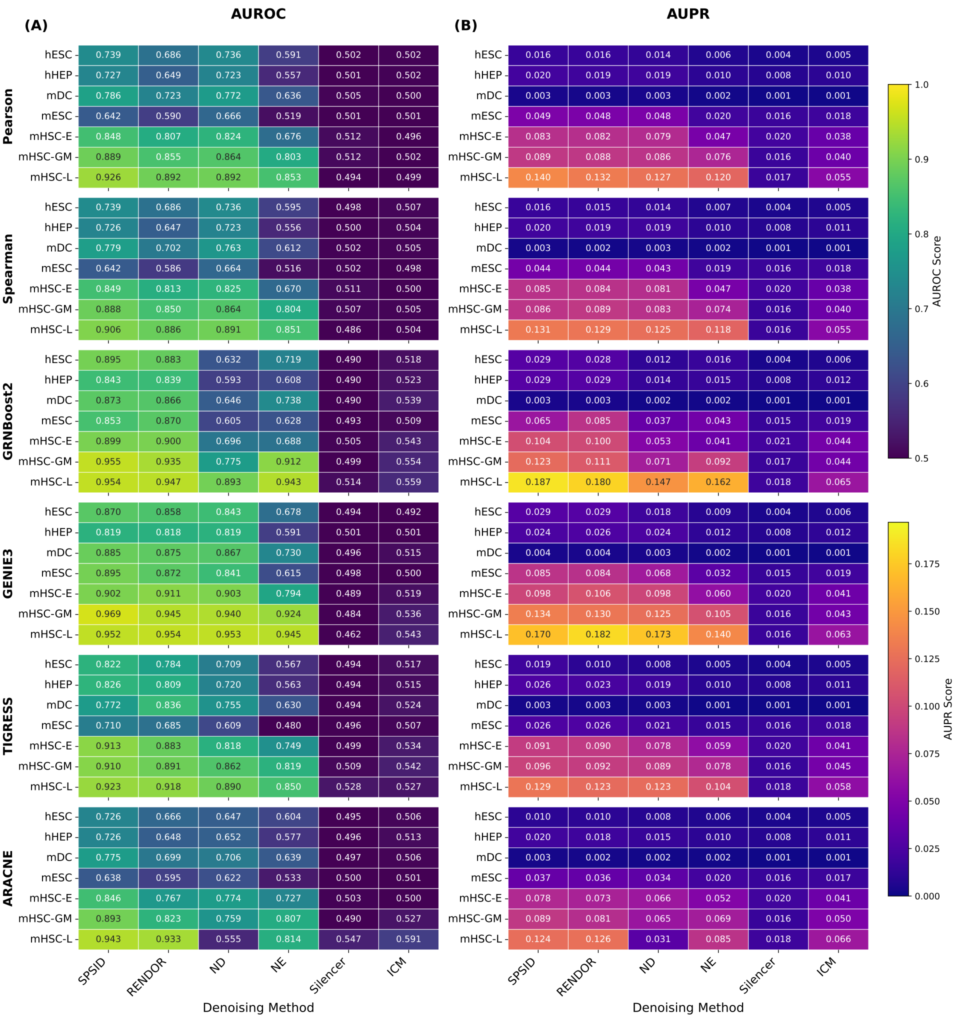


Supplementary Fig. S2: A comprehensive analysis of method ranks over the 24 DREAM5 combinations. (A) and (B) grouped bar charts display the mean rank-score for AUROC and AUPR, respectively. Error bars indicate half the inter-quartile range (IQR/2), providing a measure of variance across the six inference engines. (C) and (D) lollipop plots show the average ordinal rank determined by the Friedman test for AUROC and AUPR. In all panels, superior performance is indicated by a higher rank-score (for the bar charts) or a lower average rank (for the lollipop plots). Methods shown are SPSID (Single-Parameter Shrinkage Inverse-Diffusion), Reverse Network Diffusion On Random Walks (RENDOR), Network Deconvolution (ND), Network Enhancement (NE), Global Silencing of Indirect Correlations (Silencer), and Inverse Correlation Matrix (ICM).

Supplementary Text S3. Validation on scRNA-seq Benchmarks

To further validate the robustness and generalizability of SPSID, we evaluated it on seven scRNA-seq datasets curated by the BEELINE benchmark: mouse embryonic stem cells (mESC; GEO: GSE98664), mouse hematopoietic stem/progenitor cells with three lineage subsets (mHSC-E/GM/L; GEO: GSE81682), mouse dendritic cells (mDC; GEO: GSE48968), human embryonic stem cells (hESC; GEO: GSE75748), and human mature hepatocytes (hHEP; GEO: GSE81252). Following BEELINE's preprocessing protocol, we applied all six upstream inference methods (Pearson, Spearman, GENIE3, GRNBoost2, TIGRESS, and ARACNE), followed by SPSID and all five competing denoisers. We evaluated all results against the cell-type-specific ground truth network.

The full results are presented in Supplementary Fig. S3. SPSID was the top-performing denoising method in the vast majority of the 42 combinations (7 datasets $\times$ 6 upstream methods), achieving the highest AUROC in 37 out of 42 (88%) tasks and the highest AUPR in 34 out of 42 (81%) tasks. For instance, on the hESC-GRNBoost2 task, SPSID achieved top-performing scores for both AUROC (0.895) and AUPR (0.029). This comprehensive benchmark confirms that SPSID's denoising capability is highly effective and robust, generalizing successfully to sparse single-cell data.



Supplementary Fig. S3: scRNA-seq benchmark performance. Facet heatmap displaying AUROC (A) and AUPR (B) scores. Each row-group represents one of the 6 upstream inference methods, and each row within a group represents one of the 7 scRNA-seq datasets. Columns show SPSID and the 5 other denoising methods.

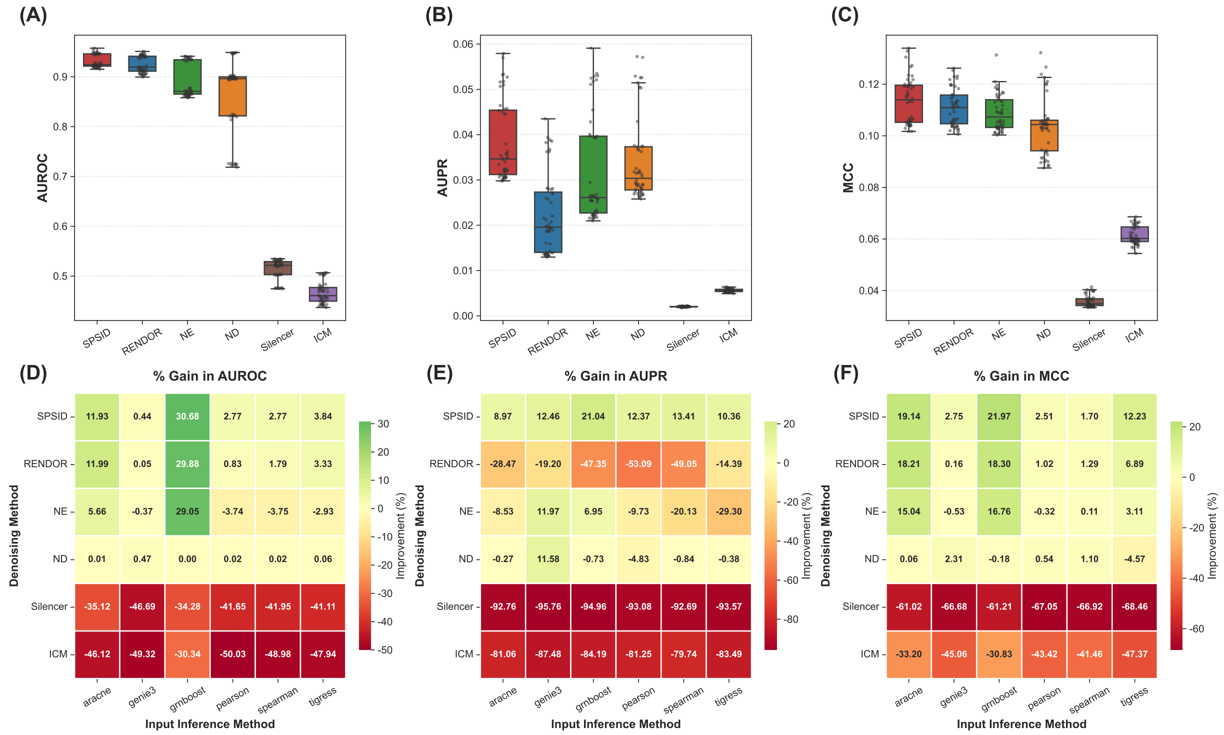
Supplementary Text S4. Sensitivity to GRN Incompleteness

To assess how sensitive SPSID is to incompleteness in regulatory annotations, we revisited the historical *E. coli* K-12 MG1655 regulatory networks curated in Abasy Atlas (v2.9; NCBI taxonomy ID 511145). Abasy provides time-stamped curation snapshots and separates interactions by evidence strength. Because we aim to test robustness to incomplete annotation under a conservative ground truth, we used the strong-evidence set (eStrong) throughout, avoiding lower-confidence associations that may mix direct regulation with indirect effects.

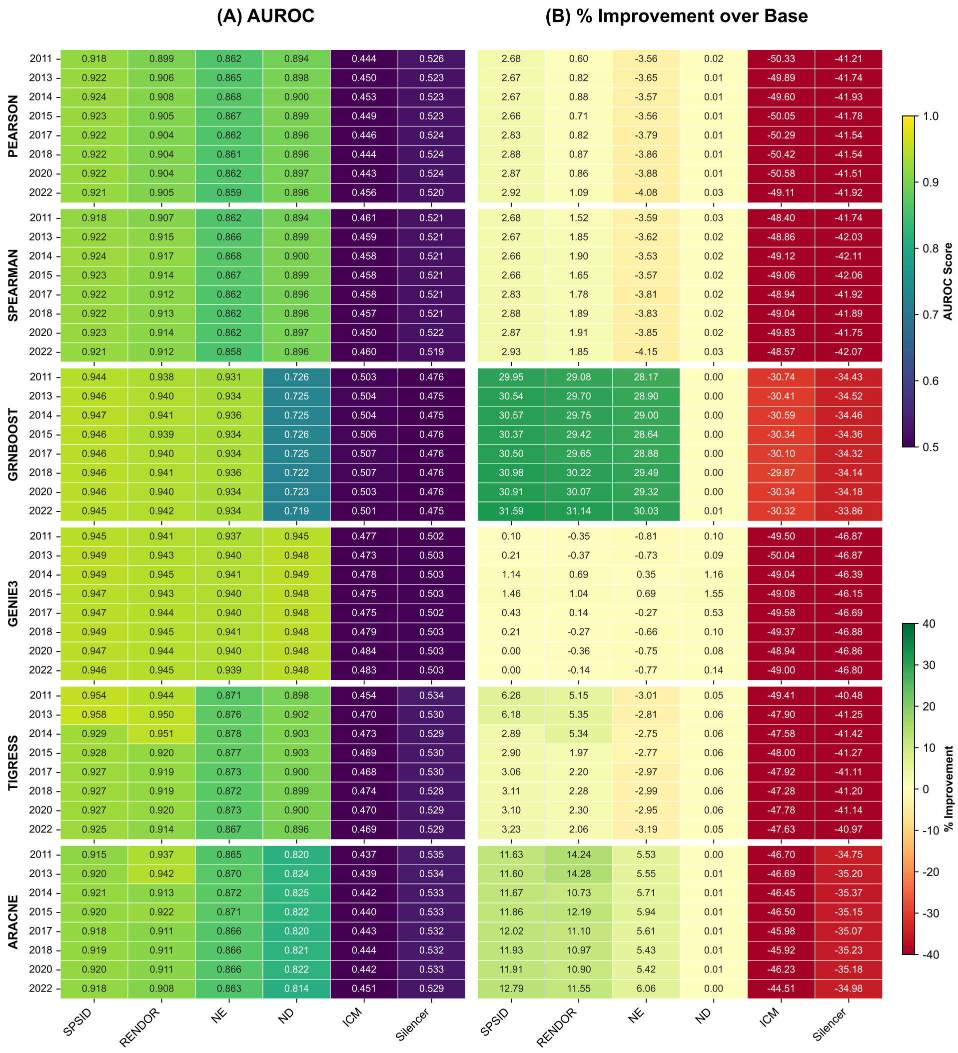
Using the official DREAM5 Network 3 mapping file, we mapped the genes evaluated in DREAM5 to Abasy gene identifiers. Under the eStrong definition, snapshots are available from 2011 onwards, and we therefore used eight curated snapshots from 2011 to 2022. The exact Abasy identifiers (taxID 511145) were:
`511145_v2011_sRDB11_eStrong`, `511145_v2013_sRDB13_eStrong`,
`511145_v2014_sRDB16_eStrong`, `511145_v2015_sRDB16_eStrong`,
`511145_v2017_sRDB16_eStrong`, `511145_v2018_sRDB19_eStrong`,
`511145_v2020_sRDB19_eStrong`, `511145_v2022_sRDB22_eStrong`.

Across snapshots, the upstream inferred networks were held fixed and identical to our DREAM5 experiments (Pearson, Spearman, GRNBoost2, GENIE3, TIGRESS, ARACNE). For each backbone and snapshot pair, we applied SPSID and all competing denoisers and evaluated AUROC, AUPR, and MCC. In addition to absolute scores, we report percentage improvement over the raw (Base) upstream prediction to make effect sizes comparable across backbones.

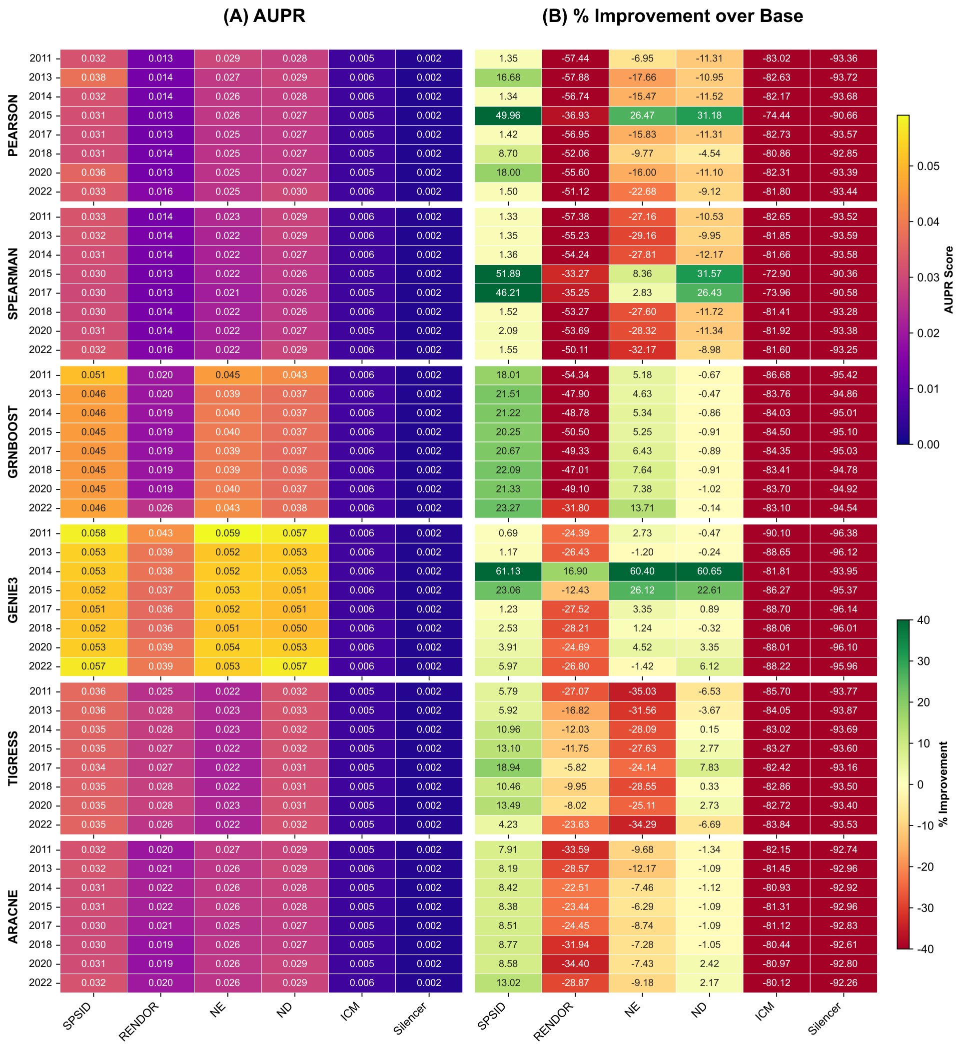
Supplementary Fig. S4 aggregates results across all eight snapshots under strong-evidence evaluation. SPSID yields consistent gains over Base across backbones and metrics. Averaged over snapshots, SPSID improves AUROC by 11.93% for ARACNE, 30.68% for GRNBoost2, and 2.77% for both Pearson and Spearman (Fig. S4D). Correspondingly, it improves AUPR by 8.97%–21.04% across backbones (Fig. S4E) and improves MCC by 1.70%–21.97% (Fig. S4F). These effect sizes are reflected in the snapshot-level heatmaps (Supplementary Figs. S5–S7). For instance, under GRNBoost2 with the 2022 eStrong snapshot, AUROC increases by 31.59% (SPSID), AUPR increases by 23.27%, and MCC increases by 23.78% (Figs. S5–S7). The strong-evidence snapshot evaluation shows that SPSID's advantages persist across multiple Abasy curation releases and across diverse upstream inference backbones. This supports that the observed improvements are not tied to a particular snapshot choice, but instead reflect robust suppression of indirect transitive structure in inferred networks under a conservative ground truth.



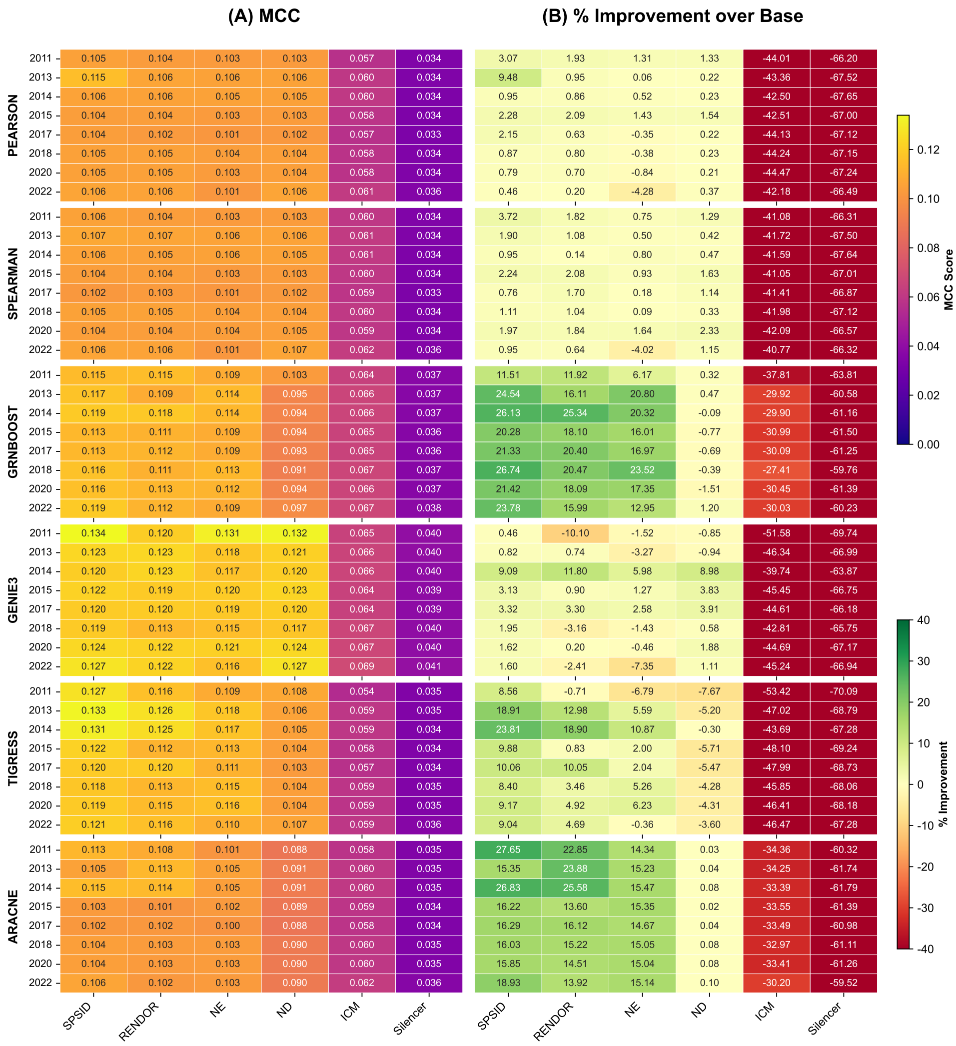
Supplementary Fig. S4: Summary of Abasy (v2.9, strong) snapshot-based evaluations. (A–C) box plots of AUROC, AUPR, and MCC across denoising methods. (D–F) average percentage gain over the raw (Base) network for AUROC, AUPR, and MCC, stratified by upstream inference backbones (ARACNE, GENIE3, GRNBoost2, Pearson, Spearman, TIGRESS).



Supplementary Fig. S5: AUROC of denoising methods as a function of Abasy *E. coli* snapshot year. Rows correspond to upstream inference methods (Pearson, Spearman, GRNBoost2, GENIE3, TIGRESS, ARACNE); columns correspond to six denoising methods. Panel (B) reports percentage improvement over Base for each denoiser.



Supplementary Fig. S6: AUPR of denoising methods as a function of Abasy *E. coli* snapshot year. Rows correspond to upstream inference methods (Pearson, Spearman, GRNBoost2, GENIE3, TIGRESS, ARACNE); columns correspond to six denoising methods. Panel (B) reports percentage improvement over Base for each denoiser.



Supplementary Fig. S7: MCC of denoising methods as a function of Abasy *E. coli* snapshot year. Rows correspond to upstream inference methods (Pearson, Spearman, GRNBoost2, GENIE3, TIGRESS, ARACNE); columns correspond to six denoising methods. Panel (B) reports percentage improvement over Base for each denoiser.